UNIVERSALLY INJECTIVE BASE CHANGE OF A FLAT FINITE PRESENTATION MODULE OVER A LOCAL HOMOMORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; the Xy-pic xy environment is required because the counted proof opens with a displayed commutative square. No macro was invented or redefined.

Editorial note. Same-chapter referenced results are located at their \label line inside the pinned chapter file; the five cross-chapter algebra lemmas are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The semi-local localisation attached to a flat finite presentation module. Same chapter, Lemma labelled lemma-homothety-spectrum, source line 1783 (Stacks tag 05DE). Supplies that Sigma^{-1}S is semi-local with maximal ideals in bijection with the associated primes of N/m_R N; the faithful flatness conclusion drawn from the hypothesis on Spec(S') -> Spec(S) is the counted step.
- Universal injectivity of the localisation map. Same chapter, Lemma labelled lemma-homothety-universally-injective, source line 1821 (Stacks tag 05DF). Supplies that N -> Sigma^{-1}N is R-universally injective; the composition argument that finishes the proof stays inside the counted span.
- Standard permanence and tensor facts for universally injective maps and faithfully flat ring maps. Algebra, Lemmas algebra-lemma-universally-injective-permanence (05CJ), algebra-lemma-ff-rings (00HQ), algebra-lemma-faithfully-flat-universally-injective (05CK), algebra-lemma-universally-injective-tensor (05CH) and algebra-lemma-composition-universally-injective (05CI). Cited at proof lines 1877-1902 for permanence under a further localisation, for faithful flatness from the condition on spectra, for universally injective maps induced by faithfully flat ring maps and for composition; none of them contains the reduction or the diagram chase of this proof.

Source reference. Original lemma 05DE: sources/stacks-flat/pinned-original-source.tex, source lines 1783–1783. Same-chapter pointer invoked at proof line 1882 for semi-locality of Sigma^{-1}S and the bijection with associated primes; the deduction of faithful flatness from the hypothesis on the image of Spec(S') -> Spec(S) is retained in the counted span.

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Source reference. Original lemma 05DF: sources/stacks-flat/pinned-original-source.tex, source lines 1821–1821. Same-chapter pointer invoked at proof line 1898 for universal injectivity of N -> Sigma^{-1}N; the composition step that turns this into the conclusion for N -> N' is retained in the counted span.

Source reference. Original lemma 05CJ: sources/stacks-chow/tags.txt, source lines 6922–6922. Cross-chapter standard prerequisite invoked once at proof lines 1877-1878 for the reduction step from N -> N' to N -> Sigma^{-1}N'.

Source reference. Original lemma 00HQ: sources/stacks-chow/tags.txt, source lines 652–652. Cross-chapter standard prerequisite invoked once at proof lines 1887-1889 for faithful flatness of a ring map whose induced map on spectra hits all maximal ideals of a semi-local ring.

Source reference. Original lemma 05CK: sources/stacks-chow/tags.txt, source lines 6923–6923. Cross-chapter standard prerequisite invoked once at proof lines 1894-1895 for universal injectivity of the module map induced by a faithfully flat ring map.

Source reference. Original lemma 05CH: sources/stacks-chow/tags.txt, source lines 6920–6920. Cross-chapter standard prerequisite invoked together with 05CK at proof lines 1895-1896 to identify the localised map with N tensor_S Sigma^{-1}S -> N tensor_S Sigma^{-1}S'.

Source reference. Original lemma 05CI: sources/stacks-chow/tags.txt, source lines 6921–6921. Cross-chapter standard prerequisite invoked once at proof lines 1900-1902 for closure of universal injectivity under composition; the final conclusion for N -> N' is drawn in the counted span.

Lemma 0.1. *Let $R \to S$ be a ring map. Let N be an S-module. Let $S \to S'$ be a ring map. Assume*

1. *$R \to S$ is a local homomorphism of local rings*
2. *S is essentially of finite presentation over R,*
3. *N is of finite presentation over S,*
4. *N is flat over R,*
5. *$S \to S'$ is flat, and*
6. *the image of $\mathrm{Spec}(S') \to \mathrm{Spec}(S)$ contains all primes $\mathfrak{q}$ of S lying over $\mathfrak{m}_R$ such that $\mathfrak{q}$ is an associated prime of $N/\mathfrak{m}_R N$.*

Then $N \to N \otimes_S S'$ is R-universally injective.

Proof. Set $N' = N \otimes_R S'$. Consider the commutative diagram

$$\begin{array}{ccc} N & \longrightarrow & N' \\ \downarrow & & \downarrow \\ \Sigma^{-1}N & \longrightarrow & \Sigma^{-1}N' \end{array}$$

where $\Sigma \subset S$ is the set of elements which are not a zerodivisor on $N/\mathfrak{m}_R N$. If we can show that the map $N \to \Sigma^{-1}N'$ is universally injective, then $N \to N'$ is too (see Algebra, Lemma 05CJ).

By Lemma 05DE the ring $\Sigma^{-1}S$ is a semi-local ring whose maximal ideals correspond to associated primes of $N/\mathfrak{m}_R N$. Hence the image of $\mathrm{Spec}(\Sigma^{-1}S') \to \mathrm{Spec}(\Sigma^{-1}S)$ contains all these maximal ideals by assumption. By Algebra, Lemma 00HQ the ring map $\Sigma^{-1}S \to \Sigma^{-1}S'$ is faithfully flat. Hence $\Sigma^{-1}N \to \Sigma^{-1}N'$, which is the map

$$N \otimes_S \Sigma^{-1}S \longrightarrow N \otimes_S \Sigma^{-1}S'$$

is universally injective, see Algebra, Lemmas 05CK and 05CH. Finally, we apply Lemma 05DF to see that $N \to \Sigma^{-1}N$ is universally injective. As the composition of universally injective module maps is universally injective (see Algebra, Lemma 05CI) we conclude that $N \to \Sigma^{-1}N'$ is universally injective and we win. $\square$